Chapter 1

Exercise 1.2.A.

Proof. The two notions can be identified. Indeed, given a group G in the usual sense, define a category with a single object X, and for each $g \in G$, define a morphism $f_g : X \to X$. Further, for two morphisms f_g, f_h, define composition by $f_g \circ f_h = f_{gh}$. This gives a 1-object groupoid, since each morphism is an isomorphism: the morphism f_g has inverse $f_{g^{-1}}$.

Conversely, given a 1-object groupoid, the set of morphisms on that one object forms a group.

On the other hand, there are clearly groupoids that aren't groups, e.g. a category with two objects and only the identity morphisms on those objects. □

Exercise 1.2.B.

Proof. The identity morphism is invertible, with itself as left- and right-inverse. The composition of two invertible morphisms is again invertible, since $(f \circ g) \circ (g^{-1} \circ f^{-1}) = \mathrm{id}$. Finally, the inverse of an invertible morphism is invertible. So, the set of invertible morphisms forms a group under composition.

Given a set X, the automorphisms of X in the category of sets is the set of bijections $f : X \to X$, i.e. the automorphism group is S_X, the symmetric group on X. The automorphisms of a vector space in Vec_k are the automorphisms in the usual sense in linear algebra.

Finally, if A and B are isomorphic objects, with an isomorphism $f : A \to B$, then for an automorphism $\phi : A \to A$, the morphism $f \circ \phi \circ f^{-1}$ is an automorphism of B, with inverse $f \circ \phi^{-1} \circ f^{-1}$. So, this association gives an isomorphism of automorphism groups. □

Exercise 1.2.C. Skipped upon recommendation
Exercise 1.2.D. Skipped upon recommendation
Exercise 1.3.A.

Proof. Let A, B both be initial objects in some category. Then, since A is initial, there is a morphism $f : A \to B$, and since B is initial, there is a morphism $g : B \to A$. Then id_A and $g \circ f$ are both morphisms $A \to A$, but since A is initial, there is a unique such morphism, so $g \circ f = \mathrm{id}_A$. Similarly, $f \circ g = \mathrm{id}_B$. So, f and g are isomorphisms. Further, this isomorphism is unique since any isomorphism $A \to B$ is firstly a morphism, and there is a unique such morphism since A is initial.

The case of final objects is identical. In fact, this proof already shows it by noting that an object is final in a category if and only if it is initial in the opposite category. □

Exercise 1.3.B.

Solution. The initial object in $Sets$ is the empty set, since there is a unique map from the empty set to any other set. The final object in $Sets$ is any set with a single element, since there is a unique map from any set to this set. Note that specifying which single-element set is irrelevant, since any two such sets are uniquely isomorphic as sets.

The initial object in $Rings$ is $\mathbb{Z}$ (recall we are only considering commutative rings with unity and morphisms that preserve unities), since there is a unique morphism given by mapping 1 to $1 \in R$ for a given ring R. The final object is the zero ring.

Top is the "same" as set: the initial object is the empty space and the final object is a single-point space, since there is a unique continuous map in each case.

Given a set S, the poset category of subsets of S has $\emptyset$ as an initial object since it is a subset of each other subset of S, and it has S as a final object. If S is a topological space and we consider the poset category of open subsets of S, the initial and final objects are the same. $\square$

Exercise 1.3.C.

Proof. In one direction, suppose $s \in S$ is a zerodivisor, so that there is some nonzero $x \in A$ with $sx = 0$. Then, since $s(1 \cdot x - 1 \cdot 0)$, we get that $x/1 = 0/1$ in $S^{-1}A$. So, x is in the kernel of the map $A \to S^{-1}A$ and it isn't injective.

Conversely, suppose this map isn't injective. Then, there is some nonzero element x in the kernel, so that $x/1 = 0/1$. I.e. there is some $s \in S$ with $s(1 \cdot x - 1 \cdot 0) = 0$, i.e. $sx = 0$, whence s is a zerodivisor. $\square$

Exercise 1.3.D.

Proof. First, note that $A \to S^{-1}A$ has the property that all elements of S are mapped to invertible elements, since for $s \in S$, the image is $s/1$, which has inverse $1/s \in S^{-1}A$. Now we want to see that it is initial with respect to this property. So, let $f : A \to B$ be a morphism with each element of S mapping to an invertible element of B. Then, we'd like to define a factorization through $S^{-1}A$. Define:

$$g(a/s) = f(a)f(s)^{-1}$$

First, note that this definition makes sense, since for each $s \in S$, $f(s)$ is invertible and so $f(s)^{-1}$ exists. Second, we want to see that it is well-defined. So, suppose $a/s = b/t$, so that there is some $r \in S$ with $r(ta - sb) = 0$. Then,

$$0 = f(r(ta - sb)) = f(r)(f(t)f(a) - f(s)f(b))$$

Since $f(r)$ is invertible, we can cancel it, and then multiplying by the inverses of $f(t)$ and $f(s)$ gives:

$$f(a)f(s)^{-1} = f(b)f(t)^{-1}$$

so that g is well-defined. Finally, we want to see it is a ring homomorphism. First, we have $g(1/1) = f(1)f(1)^{-1} = 1$. Second, if $a/s, b/t \in S^{-1}A$, then

$$g(a/s + b/t) = g((at + bs)/(st)) = f(at + bs)f(st)^{-1} = f(a)f(s)^{-1} + f(b)f(t)^{-1} = g(a/s) + g(b/t)$$

and

$$g((a/s)(b/t)) = g((ab)/(st)) = f(ab)f(st)^{-1} = f(a)f(b)f(s)^{-1}f(t)^{-1} = g(a/s)g(b/t)$$

as desired. Thus g is a ring homomorphism. Penultimately, we show it factors f. Let $\iota : A \to S^{-1}A$ be the canonical map. Then, for $a \in A$,

$$g(\iota(a)) = g(a/1) = f(a)f(1)^{-1} = f(a)$$

so that $f = g \circ \iota$ as desired. To finally finish, we need to see that g is unique. So, suppose we have another ring homomorphism $h : S^{-1}A \to B$ with $f = h \circ \iota$. Then, let $a/s \in S^{-1}A$. We have

$$h(a/s) = h((a/1)(1/s)) = h(a/1)h(1/s) = h(a/1)h(s/1)^{-1} = h(\iota(a))h(\iota(s))^{-1} = f(a)f(s)^{-1} = g(a/s)$$

so that $h = g$. This completes the proof that $S^{-1}A$ is initial among such rings.

We address the "furthermore." Let M be an $S^{-1}A$-module. Then, it is also an A-module via $a \cdot m = (a/1) \cdot m$ for $a \in A$ and $m \in M$ (restriction of scalars). For $s \in S$, multiplication by s is an isomorphism of M since it has inverse given by multiplication by $1/s$.

Conversely, suppose M is an A module such that multiplication by s is an automorphism of M for all $s \in S$. For $x \in A$, let $\varphi(x) : M \to M$ denote multiplication by x, and let $R' = \varphi(A)$ be the image of A in the endomorphism ring of M. Then R' is a commutative ring with unity, and an application of Zorn's Lemma gives a maximal subring R of the endomorphism ring of M containing R'. Then, $\varphi : A \to R$ and each element of S maps to an invertible element of R, so this factors as a map from $S^{-1}A$, i.e. we get a map from $S^{-1}A$ into the endomorphism ring of M, i.e. M is an $S^{-1}A$-module. $\square$